

ORIGINAL ARTICLE

Factors associated with the waiting time for access to specialized oral healthcare services in Brazil

Ronald Pereira Cavalcanti¹ | Rênnis Oliveira da Silva² | Petrônio José de Lima Martelli³ | José Eudes de Lorena Sobrinho⁴ | Gilberto Alfredo Pucca Júnior⁵ | Gabriela da Silveira Gaspar⁶ | Edson Hilan Gomes de Lucena⁷ 

¹Department of Collective Health, Federal University of Pernambuco, Recife, Brazil

²Graduate Program in Dentistry, Federal University of Paraíba, João Pessoa, Brazil

³Department of Social Medicine, Federal University of Pernambuco, Recife, Brazil

⁴Faculty of Medical Sciences, University of Pernambuco, Recife, Brazil

⁵Department of Dentistry, University of Brasília, Brasília, Brazil

⁶Federal University of Pernambuco, Recife, Brazil

⁷Department of Clinical and Social Dentistry, Federal University of Paraíba, João Pessoa, Brazil

Correspondence

Edson Hilan Gomes de Lucena,
Department of Clinical and Social
Dentistry, Federal University of Paraíba,
João Pessoa, Brazil, João Pessoa, Brazil.
Email: ehglucena@yahoo.com.br

Funding information

This study was supported by the Ministry of Health's National Health Fund (*Fundo Nacional de Saúde*, in Portuguese).

Abstract

Objectives: To explore the factors associated with the waiting time for access to specialized care at Dental Specialties Centers (CEO, in Portuguese), by specialty (Stomatology, Surgery, Endodontics, Patients with Special Needs and Periodontology). **Methods:** The study was a descriptive and analytic exploratory secondary analysis of data from the 2nd phase of the National Program for Improving the Access to and Quality of CEO (PMAQ-CEO, in Portuguese). All 1097 CEO in Brazil were evaluated *in loco* in 2018. Binary logistic regression was used to analyse the likelihood of users having a shorter time for assistance at CEO, by specialty.

Results: The highest and lowest median waiting times were found for endodontics (30 days) and stomatology (5 days), respectively. Smaller centres (type I CEO) had a shorter waiting list for patients with special needs (95%CI: 1.20–3.37), Endodontics (95%CI: 1.03–3.02) and Surgery (95%CI: 1.04–3.05). As for the specialties with the longest waiting list (Endodontics and Surgery), the direct route of user access to CEO was more effective than that regulated by the Healthcare System.

Conclusions: Factors related to the service, management, and to the form of relationship with primary health care influenced the waiting time for specialized care in CEO. The contact between professionals in the oral health network (primary care and secondary) was associated with a shorter waiting time, regardless of the specialty.

KEYWORDS

dental health services, health services accessibility, secondary care, waiting lists

1 | INTRODUCTION

Dental Specialties Centers (CEO, in Portuguese), created in 2003, provide a secondary level of oral health assistance in the Brazilian health system, offering free public dental care to the population. The CEO are one of the measures of the National Oral Health Policy in Brazil, which aims to ensure comprehensive oral health care, acting

as a reference for primary health care, constituting a vital factor for general health and for the population's quality of life.¹

To be certified as an CEO and receive monthly funding from the Ministry of Health, the service must fulfil specific criteria, such as deliver care in five specialties [stomatology, surgery, endodontics, care for patients with special needs (PSN) and Periodontology] and provide a minimum number of dental offices proportional to its size (Type).²

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2021 The Authors. *Community Dentistry and Oral Epidemiology* published by John Wiley & Sons Ltd.

The National Program for Improving Access to and Quality of Specialized Dental Care Centers (PMAQ-CEO, in Portuguese) was implemented to support an external assessment *in loco* of secondary oral care services. The program was designed to aid in the definition of quality parameters and certification in order to improve and expand oral healthcare actions nationwide.³

In the first phase of the PMAQ-CEO, in 2014, the waiting time for access to specialized care was identified as a dissatisfaction factor, and both regional and social inequalities were detected.^{3,4} Although national oral health policies in several countries have not defined a standard waiting time for specialist consultations,⁵ organizing the waiting list by regulating access can allow for more equitable attention.

The management of waiting times in public health is one of the purposes of access regulation, which contributes to reducing inequality while strengthening equity in care delivery. Inequality in access to public oral health services has also been experienced in other countries, such as the United States⁶ or those with public healthcare systems, such as Canada, Spain and England.^{7–9} Nearly 50 years ago, Julian Hart developed the concept of the Inverse Care Law to demonstrate that health services are more accessible to individuals with higher socioeconomic conditions and vice-versa, perpetuating or even increasing health inequality.¹⁰ This paradigm remains true in several countries up to the present day.¹¹

The regulation of access to healthcare embodies (i) the role of the State in guaranteeing the population's right to, or fair relations with, healthcare services¹²; and (ii) the public health administration, which should create opportunities, order, and qualify the access to, and delivery of, care; (iii) the regulation of access can also be related to the itinerary travelled by the user to attend health consultations.¹³ These authors conceptualized government regulation regimens (executed by policy –and decision-makers and operationalized by institutional rules and laws) into professional (executed by professionals who provide care via both institutional and independent mechanisms, through personal contacts) and lay regimes (when the user, relatives, friends and other users with the same needs employ cognitive, relational and economic resources to make government regulations more flexible and streamline their access to care). Collectively, these possibilities indicate that regulation is a field in permanent dispute, that is, a social production.

In Brazil, the regulation of access is often limited to organizing waiting lists and assistance flows.¹⁴ Given the negative consequences of waiting times on health outcomes,¹⁵ this study aimed to analyse the factors associated with the waiting time for access to specialized care at CEO in the Brazilian healthcare system by specialty (Stomatology, Surgery, Endodontics, Patients with Special Needs and Periodontology).

2 | METHODS

2.1 | Study design

This was an exploratory cross-sectional study. The secondary database from the second phase of the PMAQ-CEO's External Assessment was used, which consisted of an *in loco* survey that analysed the

conditions of access and quality of 1097 CEO distributed nationwide, regardless of their enrolment status in the program. Of these, 55 CEO were excluded from the assessment because they were undergoing renovation, were disabled, or because their managers declined participation in the external assessment. Therefore, the present census-based study considered the 1042 CEO available and accredited by Brazil's Ministry of Health in 2018.

The data used in our study corresponded to interviews with CEO managers and dentists.³ Data from the second phase of the PMAQ-CEO were provided by the Ministry of Health (<http://aps.saude.gov.br/ape/pmaq/ciclo2ceo/>).

2.2 | Study variables

The dependent variable considered in the study model was the estimated waiting time that the patient expected to be seen in each specialty in the CEO. The waiting time consisted of the time elapsed between the confirmation of the appointment until the appointment date. It was calculated for each dental specialty and dichotomized by the median, as follows: minor oral surgery (≤ 20 or > 20 days); endodontics (≤ 30 or > 30 days); PSN care (≤ 7 or > 7 days); periodontology (≤ 15 or > 15 days); and stomatology (≤ 5 or > 5 days). We chose to dichotomize the dependent variables, considering that there is no way to guarantee the accuracy of the managers' answers regarding the exact waiting time for each specialty. In addition to facilitating understanding and, as there is no cut-off point established in the literature, we used the median.

The following independent variables were considered: the size of the CEO (type I–3 dental offices; II–4 to 6 dental offices; and III–7 or more dental offices); the presence of a manager at the CEO (yes, exclusive; yes, not exclusive; no manager); the manager's work experience (< 1 year, 1–2 years, 3–4 years, 5–9 years, > 9 years); the manager's complementary training (yes, in collective health/public health, or no); the frequency of contact between the CEO and the primary care facility – PCF (always or sometimes/never); the mechanisms for appointment scheduling at CEO (only via the PCF, only via the user or via the PCF/user); referral-based access to CEO (yes or no); the percentage of absenteeism ($\leq 19\%$ or $> 19\%$); actions to reduce absenteeism (get in contact in advance/overbook or none); the existence of pre-defined allotments for the referral of users to specialized care, by specialty (yes or no); and proof of agreed clinical protocols that guide the referral of users via the PCF to CEO, by specialty (yes or no).

2.3 | Data analysis

Due to the lack of empirical and theoretical evidence regarding the waiting time for the studied dental specialties, we chose to carry out an exploratory data analysis.

The frequency distribution of variables was entered into a table according to the waiting time of each specialty. Then, a binary

logistic regression model was obtained to determine the likelihood of users having a shorter waiting time for assistance at CEO, by specialty. The independent variables were inserted into the model and submitted to adjustment using the backward stepwise procedure (Wald). Variables with $p > .20$ were removed from the adjusted model and those with $p < .05$ were considered statistically significant. Odds Ratio (OR) measures and 95% confidence intervals (CI) were obtained for the study variables. Data were tabulated and analysed in the Statistical Package for Social Sciences program (IBM-SPSS, v.24, IBM, Chicago, IL).

This study followed the ethical principles of the National Health Council's Resolution No. 416/2012 and was previously approved by the Research Ethics Committee of the Center for Health Sciences at the Federal University of Pernambuco, under protocol CAAE 23458213.0.0000.5208.

3 | RESULTS

Among the 1042 CEO, 418 (40.1%) were type I, 488 (46.8%) were type II, and 136 (13.1%) were type III. The specialties with the highest and lowest median waiting time were endodontics (30 days) and stomatology (5 days), respectively. Most CEO had a manager ($n = 994$), and in 38.7% ($n = 403$) of the cases, the manager worked exclusively in this position. The data further showed that 60% of the CEO's managers ($n = 625$) had a work experience of <2 years (Table 1).

3.1 | Stomatology

Overall, 57.7% ($n = 203$) of the CEO that had a manager with complementary training in collective/public health had a waiting time for stomatology care of less than 5 days (Table 1).

The adjusted multivariate logistic regression model indicated that CEO whose managers had over 10 years of experience in the position (OR = 2.39; 95%CI: 1.05–5.43) and those whose professionals always communicated with the primary care team (OR = 1.59; 95%CI: 1.11–2.28), were more likely to have a waiting time for stomatology care of up to 5 days (Table 2).

3.2 | Surgery

As for surgery, 60% ($n = 251$) of the CEO that contacted the primary care team had a waiting time of up to 20 days (Table 1).

Type I CEO were 1.8-fold more likely to offer specialized care in surgery within 20 days (OR = 1.78; 95%CI: 1.04–3.05). In addition, the following conditions were associated with a greater likelihood that surgical care would be delivered within 20 days: (i) CEO that always contacted the PCF (OR = 1.57; 95%CI: 1.12–2.20); those that allowed scheduling by users only (OR = 2.60; 95%CI: 1.65–4.10) or via the PCF and/or user (OR = 1.86; 95%CI: 1.15–3.02); those that were not limited exclusively to referred patients (OR = 1.57; 95%CI:

1.10–2.25); CEO with a percentage of absenteeism lower than 19% (OR = 1.49; 95%CI: 1.06–2.10); and CEO with allotments for surgery previously set up by the primary oral healthcare team (OR = 1.59; 95%CI: 1.05–2.43) (Table 2).

3.3 | Endodontics

For endodontics, 62.2% ($n = 227$) of the CEO whose managers had work experience of 1 and 2 years in the position had their waiting time shorter than 30 days (Table 1).

As for specialized care in endodontics, the following conditions were associated with a greater likelihood that care would be delivered within 30 days: type I CEO (OR = 1.76; 95%CI: 1.03–3.02); CEO that always communicated with the PCF (OR = 1.51; 95%CI: 1.07–2.13); those that allowed scheduling only by users (OR = 3.11; 95%CI: 1.97–4.93) or via the PCF and/or users (OR = 2.36; 95%CI: 1.45–3.84); those that were not limited exclusively to referred patients (OR = 1.58; 95%CI: 1.10–2.27); CEO with a percentage of absenteeism lower than 19% (OR = 1.50; 95%CI: 1.07–2.12); and CEO with endodontic appointments previously set up by the primary oral care team (OR = 1.82; 95%CI: 1.21–2.75) (Table 2).

3.4 | Patients with Special Needs (PSN)

For the specialty of PNS, 61.3% ($n = 317$) of the CEO that contacted the user in advance or 'overbooked' the appointment list had a waiting time shorter than 7 days (Table 1).

Type I CEO (OR = 2.01; 95%CI: 1.20–3.37) that contacted users in advance or 'overbooked' the appointment list (OR = 1.91; 95%CI: 1.31–2.77) and those that always communicated with the PCF (OR = 1.86; 95%CI: 1.33–2.61) were more likely to have a waiting time for PSN care within 7 days (Table 2).

3.5 | Periodontology

As for periodontology, 72.2% ($n = 273$) of the CEO that did not limit their appointments to referred-only demands showed a waiting time shorter than 15 days (Table 1).

As for specialized periodontal care, CEO that always contacted the primary care team were 1.5-fold (OR = 1.53; 95%CI: 1.09–2.15) more likely to have a waiting time of up to 15 days. CEO that allowed scheduling only by users (OR = 2.21; 95%CI: 1.47–3.32) or via the PCF and/or users (OR = 1.69; 95%CI: 1.08–2.62) were also more likely to deliver periodontal care within 15 days (Table 2).

4 | DISCUSSION

The findings indicated that the median waiting time for access to specialized dental care in the Brazilian healthcare system ranged from 5 to

TABLE 1 Absolute and relative frequency distribution of the independent variables according to the waiting time for specialized care (stomatology, surgery, endodontics, care for patients with special needs, periodontology) at CEO.

Waiting time													
Variables		Stomatology			Surgery		Endodontics		PSN		Periodontology		
		Total of CEO	>5 days	≤5 days	>20 days	≤20 days	>30 days	≤30 days	>7 days	≤7 days	>15 days	≤15 days	
Size of the CEO (Type)													
I	418 (40.1)	155 (47.8)	169 (52.2)	148 (36.3)	260 (63.7)	159 (39.5)	244 (60.5)	136 (35.1)	251 (64.9)	118 (30)	275 (70)		
II	488 (46.8)	202 (49.5)	206 (50.5)	243 (52.9)	216 (47.1)	230 (50.9)	222 (49.1)	209 (46.1)	244 (53.9)	166 (36.9)	284 (63.1)		
III	136 (13.1)	47 (41.2)	67 (58.8)	69 (57)	52 (43)	64 (52.5)	58 (47.5)	58 (48.3)	62 (51.7)	47 (39.2)	73 (60.8)		
Presence of a manager at the CEO													
Yes, exclusive.	403 (38.7)	142 (43.4)	185 (56.6)	192 (51.6)	180 (48.4)	185 (50.3)	183 (49.7)	169 (46.8)	192 (53.2)	136 (37.5)	227 (62.5)		
Yes, not exclusive.	591 (56.7)	247 (50.3)	244 (49.7)	246 (42.9)	327 (57.1)	243 (42.9)	323 (57.1)	221 (39.7)	336 (60.3)	182 (32.7)	375 (67.3)		
No manager.	48 (4.6)	15 (53.6)	13 (46.4)	22 (51.2)	21 (48.8)	25 (58.1)	18 (41.9)	13 (31)	29 (69)	13 (30.2)	30 (69.8)		
The manager's work experience													
<1 year	245 (23.5)	99 (49.7)	100 (50.3)	105 (44.7)	130 (55.3)	112 (48.9)	117 (51.1)	93 (40.6)	136 (59.4)	85 (38.1)	138 (61.9)		
1-2 years	380 (36.5)	162 (51.6)	152 (48.4)	154 (41.8)	214 (58.2)	138 (37.8)	227 (62.2)	148 (41)	213 (59)	105 (29)	257 (71)		
3-4 years	107 (10.3)	40 (47.1)	45 (52.9)	50 (51.5)	47 (48.5)	45 (46.4)	52 (53.6)	45 (47.9)	49 (52.1)	38 (38.8)	60 (61.2)		
5-9 years	166 (15.9)	59 (40.1)	88 (59.9)	85 (54.8)	70 (45.2)	86 (56.2)	67 (43.8)	63 (42)	87 (58)	59 (38.6)	94 (61.4)		
More than 9 years	49 (4.7)	16 (38.1)	26 (61.9)	24 (52.2)	22 (47.8)	28 (60.9)	18 (39.1)	22 (51.2)	21 (48.8)	14 (33.3)	28 (66.7)		
The manager's complementary training													
Yes, in Public Health	402 (38.6)	149 (42.3)	203 (57.7)	195 (51.9)	181 (48.1)	187 (50)	187 (50)	161 (43)	213 (57)	146 (39)	228 (61)		
No	592 (56.8)	240 (51.5)	226 (48.5)	243 (42.7)	326 (57.3)	241 (43)	319 (57)	229 (42.1)	315 (57.9)	172 (31.5)	374 (68.5)		
Frequency of contact between CEO and PCF													
Always	432 (41.5)	159 (44.3)	200 (55.7)	167 (40)	251 (60)	170 (41)	245 (59)	138 (33.9)	269 (66.1)	115 (28.4)	290 (71.6)		
Sometimes/never	665 (63.8)	245 (50.3)	242 (49.7)	293 (51.4)	277 (48.6)	283 (50.4)	279 (49.6)	265 (47.9)	288 (52.1)	216 (38.7)	342 (61.3)		
Mechanisms for appointment scheduling at CEO													
Only via the PCF	215 (20.6)	78 (42.6)	105 (57.4)	127 (67.6)	61 (32.4)	123 (66.5)	62 (33.5)	83 (43.5)	108 (56.5)	93 (50.3)	92 (49.7)		
Only via the user	566 (54.3)	228 (51)	219 (49)	212 (38.8)	334 (61.2)	215 (39.8)	325 (60.2)	215 (40.8)	312 (59.2)	154 (28.9)	378 (71.1)		
Via the PCF/user	316 (30.3)	98 (45.4)	118 (54.6)	121 (47.6)	133 (52.4)	115 (45.6)	137 (54.4)	105 (43.4)	137 (56.6)	84 (34.1)	162 (65.9)		
Referral-based access to CEO													
Yes	637 (61.1)	243 (46.2)	283 (53.8)	312 (52.4)	283 (47.6)	304 (51.5)	286 (48.5)	241 (41.4)	341 (58.6)	226 (38.6)	359 (61.4)		
No	405 (38.9)	161 (50.3)	159 (49.7)	148 (37.7)	245 (62.3)	149 (38.5)	238 (61.5)	162 (42.9)	216 (57.1)	105 (27.8)	273 (72.2)		

TABLE 1 (Continued)

Variables		Waiting time									
		Stomatology					Surgery				
		Total of CEO		≤5 days		≤20 days		>30 days		PSN	
		N (%)	N (%)	N (%)	N (%)	N (%)	N (%)	N (%)	N (%)	≤7 days	≤15 days
										N (%)	N (%)
Percentage of absenteeism											
≤19%		392 (37.6)	147 (44.7)	182 (55.3)	167 (44.1)	212 (55.9)	163 (43)	216 (57)	145 (38.7)	230 (61.3)	245 (65.7)
>19%		380 (36.5)	149 (45.7)	177 (54.3)	208 (59.1)	144 (40.9)	201 (57.8)	147 (42.2)	161 (46.7)	184 (53.3)	214 (62)
Actions to reduce absenteeism											
Get in contact in advance/overbook		549 (52.7)	207 (43.3)	271 (56.7)	262 (50.3)	259 (49.7)	252 (48.6)	267 (51.4)	200 (38.7)	317 (61.3)	338 (65.9)
None		223 (21.4)	89 (50.3)	88 (49.7)	113 (53.8)	97 (46.2)	112 (53.8)	96 (46.2)	106 (52.2)	97 (47.8)	121 (59)
Allotments for surgery											
Yes		205 (19.7)	69 (43.1)	91 (56.9)	92 (48.4)	98 (51.6)	89 (47.8)	97 (52.2)	80 (43.5)	104 (56.5)	107 (58.5)
No		823 (79.0)	331 (48.9)	346 (51.1)	366 (46.3)	425 (53.7)	359 (46)	421 (54)	319 (41.6)	447 (58.4)	518 (67.3)
Allotments for endodontics											
Yes		230 (22.1)	80 (42.8)	107 (57.2)	100 (47.2)	112 (52.8)	95 (45.2)	115 (54.8)	87 (42.2)	119 (57.8)	127 (61.4)
No		795 (76.3)	319 (49.2)	329 (50.8)	357 (46.9)	405 (53.1)	357 (46.9)	405 (53.1)	313 (42.2)	428 (57.8)	497 (66.7)
Allotments for PSN											
Yes		174 (16.7)	63 (47)	71 (53)	72 (45.6)	86 (54.4)	71 (45.8)	84 (54.2)	74 (47.1)	83 (52.9)	90 (58.8)
No		821 (78.8)	334 (49)	348 (51)	370 (47)	417 (53)	369 (47.3)	411 (52.7)	328 (41.8)	456 (58.2)	520 (67.4)
Allotments for Periodontology											
Yes		189 (18.1)	65 (43.6)	84 (56.4)	85 (49.1)	88 (50.9)	81 (47.9)	88 (52.1)	73 (43.7)	94 (56.3)	97 (57.4)
No		823 (79.0)	333 (48.8)	349 (51.2)	365 (46.3)	424 (53.7)	362 (46.2)	421 (53.8)	322 (41.8)	449 (58.2)	526 (67)
Allotments for stomatology											
Yes		108 (10.4)	56 (55.4)	45 (44.6)	49 (48.5)	52 (51.5)	48 (48.5)	51 (51.5)	41 (41.4)	58 (58.6)	54 (54.5)
No		745 (71.5)	343 (49.8)	346 (50.2)	354 (49.8)	357 (50.2)	344 (48.9)	359 (51.1)	300 (42.7)	402 (57.3)	453 (64.6)
Proof of protocols for surgery											
Yes		792 (76.0)	308 (45.6)	367 (54.4)	380 (50.5)	373 (49.5)	367 (49.1)	381 (50.9)	316 (42.4)	430 (57.6)	465 (63)
No		95 (9.1)	39 (54.2)	33 (45.8)	29 (31.2)	64 (68.8)	34 (37)	58 (63)	32 (36)	57 (64)	66 (74.2)
Proof of protocols for endodontics											
Yes		799 (76.7)	311 (45.8)	368 (54.2)	385 (50.5)	377 (49.5)	373 (49.3)	384 (50.7)	322 (42.8)	431 (57.2)	469 (63)
No		94 (9.0)	40 (56.3)	31 (43.7)	26 (28.9)	64 (71.1)	32 (35.2)	59 (64.8)	30 (35.3)	55 (64.7)	65 (73.9)

(Continues)

TABLE 1 (Continued)

Variables	Waiting time									
	Stomatology		Surgery		Endodontics		PSN		Periodontology	
	>5 days N (%)	≤5 days N (%)	>20 days N (%)	≤20 days N (%)	>30 days N (%)	≤30 days N (%)	>7 days N (%)	≤7 days N (%)	>15 days N (%)	≤15 days N (%)
Proof of protocols for PSN										
Yes	771 (74.0)	305 (46.6)	350 (53.4)	360 (49)	356 (48.8)	373 (51.2)	320 (43.8)	411 (56.2)	270 (37.6)	449 (62.4)
No	78 (7.5)	34 (55.7)	27 (44.3)	53 (68.8)	31 (40.3)	46 (59.7)	31 (40.8)	45 (59.2)	19 (25.7)	55 (74.3)
Proof of protocols for periodontology										
Yes	791 (75.9)	313 (46.4)	362 (53.6)	372 (49.4)	366 (48.9)	382 (51.1)	319 (42.8)	426 (57.2)	277 (37.4)	463 (62.6)
No	86 (8.3)	37 (56.1)	29 (43.9)	61 (72.6)	32 (38.1)	52 (61.9)	29 (36.3)	51 (63.8)	22 (26.8)	60 (73.2)
Proof of protocols for stomatology										
Yes	652 (62.6)	290 (48.2)	312 (51.8)	283 (45.6)	310 (50.4)	305 (49.6)	268 (43.4)	350 (56.6)	237 (38.9)	373 (61.1)
No	72 (6.9)	37 (54.4)	31 (45.6)	48 (67.6)	33 (46.5)	38 (53.5)	27 (38.6)	43 (61.4)	22 (31.4)	48 (68.6)

Note: The data were obtained from PMAQ-CEO 2nd phase of external assessments, Brazil, 2018.

Abbreviation: CEO, Dental Specialties Centers; PCF, Primary Care Facility; PSN, Patients with Special Needs.

30 days across specialties. The longest waiting list was for endodontic care, which is consistent with other reports from different years and locations in Brazil.¹⁶⁻¹⁸ This is particularly relevant in the context of national oral health policies since initiatives to reduce the waiting list can directly impact health outcomes nationwide. While the literature has shown inarguably that long waiting times are associated with increased absenteeism, high costs/economic burden, and health issues,^{19,20} this remains poorly debated in the field of oral health.²¹

The frequency of contact between professionals working at CEO and primary care services proved to be a significant mechanism for regulating the access to specialized care when compared with the user referral modality. For all dental specialties, communication inside the healthcare network increased by 1.5-fold the likelihood of reducing the median waiting time for specialized dental care, varying between the different specialties from 1.51 (endodontics) to 1.86 (PSN). However, we note that the contact between professionals should not imply the flexibilization of formal regulation nor should it be seen as an independent initiative, which could negatively open room for personal networking between the services. As it stands, this type of initiative may cause inequity of access to care and favour specific groups of users.¹⁴

The regulation of access through computerized appointment scheduling via the primary care team, and/or the operationalization of CEO based exclusively on referrals, may create barriers and additional bureaucracy in the flow of users. This could produce a vertical, normative and centralized regulation that does not consider complementary actions of dialogues and agreements within the healthcare network.¹⁴

Here, our findings indicated that when the search for specialized care in surgery, endodontics and periodontology was led by the user, the waiting time for consultation was significantly reduced. This can be understood as a 'lay regulation', in which the user, relatives, friends and other users with the same needs make use of cognitive, relational and economic resources to make government regulations more flexible and streamline their access to specialized care.¹⁴ About the first phase of the PMAQ-CEO,⁴ these findings demonstrate that the regulation of access based on a State action ('governmental regulation') made little progress in the organization of user flow.

The absence of referral from primary health care can compromise the functioning of the oral health network. A study carried out in a metropolitan region in northeastern Brazil observed that when referred to the CEO by the primary healthcare dentist, the user's chance of returning to this level of care is considerably higher than the absence of this reference, thus contributing to the comprehensiveness in oral health care. In addition, free demand can compromise the performance of CEO when they are used for low-complexity care.²²

The percentage of absenteeism lower than 19% was a decisive factor for reducing the waiting time for endodontic and surgery care—which were the specialties with the longest waiting list. In this case, the existence of pre-defined allotments for the referral of users to specialized care increased the probability of having shorter waiting times. Consistent with this, long waiting times and absenteeism were previously regarded in the literature as mutually enhancing factors.¹⁸ In addition, these long queues can produce, in users,

care needs that were not initially planned and that exceed the pre-defined limits of referral for specialized care.

Inequalities in the waiting time for access to specialized care at CEO have been evidenced in Brazil since the first phase of the PMAQ-CEO in 2014. On that occasion, the user's profile that had the best access to specialized care, that is, the shortest waiting time, was the one with the highest family income and schooling.⁴ According to a previous study, user dissatisfaction was greater in the North and Northeast regions of Brazil; as well as among those who travelled large distances to have access to specialized care; and among those who had to wait more than one month for the first appointment.⁵

Our data showed that a lower percentage of absenteeism—a factor known for increasing waiting time—had a positive impact on access to the most demanded specialties, such as endodontics and surgery. As for the PNS specialty, actions such as prior contact and overbooking, which aimed to reduce absenteeism, were also associated with a shorter waiting time for appointments at the CEO. These are positive factors, especially if we consider all the difficulties that the PNS patient has to go to health services. These findings add to the current knowledge about care with PSN, which points to the need to delve deeper into the possible influence of service delivery models and barriers to accessing care for individuals with special needs and to ensure equitable access to health care.²³

TABLE 2 Adjusted multivariate logistic regression model to determine the likelihood of CEO having a shorter waiting time for providing specialized care (Stomatology, Surgery, Endodontics, Patients with special needs and Periodontology), according to PMAQ-CEO 2nd phase of external assessments, Brazil, 2018

Speciality	Adjusted Model	OR (95% CI)
Stomatology	Presence exclusive of a manager at the CEO	
	Yes, exclusive.	1.36 (0.95–1.95)
	Yes, not exclusive.	1
	The manager's work experience	
	Less than 1 year	1
	1–2 years	0.99 (0.63–1.56)
	3–4 years	0.77 (0.40–1.45)
	5–9 years	1.54 (0.91–2.60)
	More than 9 years	2.39 (1.05–5.43)
	Frequency of contact between CEO and PCF	
	Always	1.59 (1.11–2.28)
	Sometimes/never	1
Surgery	Size of the CEO (Type)	
	I	1.78 (1.04–3.05)
	II	1.05 (0.63–1.75)
	III	1
	Frequency of contact between CEO and PCF	
	Always	1.57 (1.12–2.20)
	Sometimes/never	1
	Mechanisms for appointment scheduling at CEO	
	Only via PCF	1
	Only via user	2.60 (1.65–4.10)
	Via the PCF/user	1.86 (1.15–3.02)
	Referral-based access to CEO	
	Yes	1
	No	1.57 (1.10–2.25)
	Percentage of absenteeism	
	≤19	1.49 (1.06–2.10)
	>19	1
	Allotments for surgery	
	Yes	1.59 (1.05–2.43)
	No	1

(Continues)

TABLE 2 (Continued)

Speciality	Adjusted Model	OR (95% CI)
Endodontics	Size of the CEO (Type)	
	I	1.76 (1.03–3.02)
	II	1.05 (0.63–1.76)
	III	1
	The manager's complementary training	
	Yes, in public Health	0.74 (0.53–1.04)
	No	1
	Frequency of contact between CEO and PCF	
	Always	1.51 (1.07–2.13)
	Sometimes/never	1
	Mechanisms for appointment scheduling at CEO	
	Only via PCF	1
	Only via user	3.11 (1.97–4.93)
	Via the PCF/user	2.36 (1.45–3.84)
	Referral-based access to CEO	
	Yes	1
	No	1.58 (1.10–2.27)
	Percentage of absenteeism	
	≤ 19%	1.50 (1.07–2.12)
	> 19%	1
	Allotments for endodontics	
	Yes	1.82 (1.21–2.75)
	No	1
PSN	Size of the CEO (Type)	
	I	2.01 (1.20–3.37)
	II	1.25 (0.77–2.04)
	III	1
	Frequency of contact between CEO and PCF	
	Always	1.86 (1.33–2.61)
	Sometimes/never	1
	Actions to reduce absenteeism	
	Get in contact in advance/overbook	1.91 (1.32–2.77)
	None	1
Periodontology	Frequency of contact between CEO and PCF	
	Always	1.53 (1.09–2.15)
	Sometimes/never	1
	Mechanisms for appointment scheduling at CEO	
	Only via PCF	1
	Only via user	2.21 (1.47–3.32)
	Via the PCF/user	1.69 (1.08–2.62)

Abbreviation: CEO, Dental Specialties Centers; PCF, Primary care facility; PSN, Patients with special needs; OR, Odds ratio; 95% CI: 95% Confidence interval.

The waiting time for access to specialized dental care was found to be influenced by several factors. The contact between the primary care and CEO teams was associated with a shorter waiting time, regardless of the specialty. Furthermore, in some specialties (surgery,

endodontics and PSN), the size of the CEO was also associated with a shorter waiting time for access to specialized care. This may be because they are in cities with a smaller population size and thus must serve a smaller number of people than type III CEO, that is, larger ones.

The analysis resulted in five models, one for each specialty offered in the CEO. This is probably because each specialty has a different dynamic from daily care, and because the demands and needs of the population are different by specialty. As such, this results in a specific waiting time for each specialty even if it is in the same CEO. Given this, we could not include different specialties within the same regression model, as factors influenced waiting time differently for each specialty.

The cross-sectional methodological design has limited ability to estimate cause-and-effect relationships. In addition, data collected from the external evaluation was self-reported, which can, in some cases, be considered a source of bias. In general, the population-based study provides evidence on factors that can reduce the waiting time by specialty for users to be seen in special care in Brazil. This can serve as a basis for other longitudinal studies that include user data in the assessment.

To conclude, different factors were associated with waiting times for specialized care CEO. These factors differed according to specialty. In general, the shortest waiting time was associated with service characteristics, such as size; the characteristics of the management, such as the presence of a manager acting exclusively in the management of the service; how and how often the specialized service was related to primary health care; and actions taken to reduce absenteeism. These findings represent a challenge for the proper implementation of the regulation of access to specialized care in oral health.

ACKNOWLEDGEMENT

The authors would like to thank the Collaborating Center of the Ministry of Health at the Federal University of Pernambuco—CECOL/UFPE.

ORCID

Edson Hilan Gomes de Lucena  <https://orcid.org/0000-0003-3431-115X>

REFERENCES

1. Brasil. Ministério da Saúde. Passo a passo das ações da Política Nacional de Saúde Bucal. Brasília (DF). 2016. Available from: http://189.28.128.100/dab/docs/portaldab/publicacoes/acoes_politica_saude_bucal.pdf
2. Goes PSA, Figueiredo N, Martelli PJL, et al. Theoretical and methodological aspects of the external evaluation of the improvement, access and quality of centers for dental specialties program. *Pesq Bras Odontoped Clin Integr*. 2018;18(1):e3433.
3. Goes PSA, Cavalcanti RP, Cruz DF, Almeida LFD, Padilha WWN. The regulation of services in secondary oral health care in Brazil. *Pesq Bras Odontoped Clin Integr*. 2018;18(1):e3537.
4. Andrade FB, Pinto RS. Fatores associados à insatisfação dos usuários dos centros de especialidades odontológicas do Brasil em 2014: estudo transversal. *Epidemiologia E Serviços De Saúde*. 2020;29(3):e2019429.
5. NHS England. Achieving World-Class Cancer Outcomes: Taking the strategy forward. 2016;1-13. Available from <https://www.england.nhs.uk/wp-content/uploads/2016/05/cancer-strategy.pdf>
6. Hung M, Lipsky MS, Moffat R, et al. Health and dental care expenditures in the United States from 1996 to 2016. *PLoS One*. 2020;15(6):e0234459.
7. Dehmoobadsharifabadi A, Singhal S, Quinonez C. Investigating the “inverse care law” in dental care: A comparative analysis of Canadian jurisdictions. *Can J Public Health*. 2017;107(6):e538-e544.
8. Capurro DA, Davidsen M. Socioeconomic inequalities in dental health among middle-aged adults and the role of behavioral and psychosocial factors: evidence from the Spanish National Health Survey. *Int J Equity Health*. 2017;16(1):34.
9. Watt RG, Williams DM, Sheiham A. The role of the dental team in promoting health equity. *Br Dent J*. 2014;216(1):11-14.
10. Hart JT. The inverse care law. *Lancet*. 1971;1(7696):405-412.
11. Li C, Yao NA, Yin A. Disparities in dental healthcare utilization in China. *Community Dent Oral Epidemiol*. 2018;46(6):576-585.
12. Bahia L. Padrões e mudanças no financiamento e regulação do Sistema de Saúde Brasileiro: impactos sobre as relações entre o público e privado. *Saúde E Sociedade*. 2005;14(2):9-30.
13. Cecilio LCO, Carapinheiro G, Andreazza R, et al. O agir leigo e o cuidado em saúde: a produção de mapas de cuidado. *Cad. Saúde Pública*. 2014;30(7):1502-1514.
14. Freire MP, Louvison M, Feuerweker LCM, Chioro A, Bertussi D. Regulação do cuidado em redes de atenção: importância de novos arranjos tecnológicos. *Saúde Soc*. 2020;29(3):e190682.
15. Mundi N, Theurer J, Warner A, et al. The impact of seasonal operating room closures on wait times for oral cancer surgery. *Curr Oncol*. 2018;25(1):67-72.
16. Magalhães MBP, Oliveira DV, Lima RF, Ferreira EF, Martins RC. Avaliação da atenção secundária em endodontia em um Centro de Especialidades Odontológicas (CEO). *Cien Saude Colet*. 2019;24(12):4643-4654.
17. Laroque MB, Fassa ACG, Castilhos ED. Avaliação da atenção secundária em saúde bucal do Centro de Especialidades Odontológicas de Pelotas, Rio Grande do Sul, 2012–2013. *Epidemiol. Serv. Saúde*. 2015;24(3):421-430.
18. Machado AT, Werneck MA, Lucas SD, Abreu MH. Who did not appear? First dental visit absences in secondary care in a major Brazilian city: a cross-sectional study. *Cien Saude Colet*. 2015;20(1):289-298.
19. Conill EM, Giovanella L, Almeida PF. Listas de espera em sistemas públicos: da expansão da oferta para um acesso oportuno? Considerações a partir do Sistema Nacional de Saúde espanhol. *Ciênc. Saúde Colet*. 2011;16(6):2783-2794.
20. Farias CML, Giovanella L, Oliveira AE, Santos Neto ET. Tempo de espera e absenteísmo na atenção especializada: um desafio para os sistemas universais de saúde. *Saúde Em Debate*. 2019;43(spe5):190-204.
21. Pinto Vde P, Teixeira AH, Santos PR, de Araújo MW, Moreira MÁ, Saraiva SR. Avaliação da acessibilidade ao Centro de Especialidades Odontológicas de abrangência macrorregional de Sobral, Ceará, Brasil. *Cien Saude Colet*. 2014;19(7):2235-2244.
22. Souza GC, Lopes MLDS, Roncalli AG, Medeiros-Júnior A, Clara-Costa IC. Referência e contra referência em saúde bucal: regulação do acesso aos centros de especialidades odontológicas. *Rev. Salud Pública*. 2015;17(3):416-428.
23. Lim MAWT, Liberali SAC, Borromeo GL. Utilisation of dental services for people with special health care needs in Australia. *BMC Oral Health*. 2020;20(1):360.

How to cite this article: Cavalcanti RP, da Silva RO, Martelli PJDL, et al. Factors associated with the waiting time for access to specialized oral healthcare services in Brazil. *Community Dent Oral Epidemiol*. 2022;50:58–66. doi:[10.1111/cdoe.12720](https://doi.org/10.1111/cdoe.12720)